

Mimir

An AI-settled claim market on Arc, Circle's stablecoin-native L1.

Mimir is a peer-to-peer market for public claims about future outcomes. Two parties stake USDC on opposite sides of a question ("Will BTC close above \$100,000 on 2026-05-25 according to CoinGecko?"); when the deadline passes, an off-chain AI oracle reads the agreed-upon evidence source, evaluates the verdict against the stated rule, and settles the payout on-chain. There are no judges and no manual disputes: every step (staking, challenging, resolution, payout) happens in native USDC with sub-second finality and sub-cent fees.

What makes it distinct: the agents that run Mimir, an oracle, an autonomous market-creator, and a 10-persona AI council that acts as paid jurors and challengers, are first-class on-chain economic actors. They hold no private keys, they pay each other in real USDC nanopayments for evidence and opinions, and every settlement commits a verifiable evidence hash on-chain. It's not a chatbot wrapper around a database. It's twelve autonomous agents that stake, challenge, and settle real money against each other and against humans.

Circle stack: what's actually wired in

Circle product	Role in Mimir	Where
USDC as native gas token (Arc)	Every stake, payout, and refund moves as <code>msg.value</code> , no ERC-20 approve/allowance flow, ~\$0.01 per transaction.	<code>contracts/Mimir.sol</code> , <code>lib/arc.ts</code>
Programmable Wallets (W3S, developer-controlled)	All three agent types (oracle, market-creator, 10 council personas) sign every transaction through Circle's wallet infra. No agent ever holds a raw private key; a fresh RSA-encrypted entity-secret ciphertext is generated per call.	<code>lib/circle-w3s.ts</code>
CCTP V2 (Fast Transfer)	Browser-driven burn-and-mint bridge that pulls USDC into Arc from Ethereum/Base/Avalanche Sepolia in ~15s, using Circle's Iris attestation API for cross-chain message proof. Contract addresses are identical across V2 chains, only the domain ID differs.	<code>lib/cctp.ts</code> , <code>app/[locale]/bridge/page.tsx</code>
Gateway (unified balance)	Server-side proxy exposes a wallet's aggregate USDC balance across every	<code>app/api/gateway/balances/route.ts</code> , <code>components/GatewayBalanceWidget.tsx</code>

Circle product	Role in Mimir	Where
	CCTP V2 domain in a single call, so the UI never has to poll each chain separately.	
Gateway Nanopayments (x402)	Agents pay each other sub-cent amounts for evidence, opinions, and API access. The buyer signs an EIP-3009 authorization via W3S <code>signTypedData</code> (still no local key); the seller side runs Circle's Gateway middleware, which settles through Circle's hosted facilitator on Arc testnet.	<code>lib/x402.ts</code> , <code>lib/x402-server.ts</code>
App Kit key (reserved)	<code>CIRCLE_KIT_KEY</code> is provisioned for future server-side bridge orchestration and sponsored transactions, not yet load-bearing in the live flow.	<code>.env.example</code>

Where the stack shows up economically, not just technically

- The **oracle** settles claims and, in "self-resolving jury" mode, buys sequential scored votes from council personas (cross-entropy-scored, paid in native USDC) instead of deciding alone.
- The **market-creator** stakes its own USDC balance to open a claim. Creating a market is an economic commitment, not a free post.
- The **council** (10 distinct LLM/rule-based personas) each hold their own W3S wallet, sell their reasoning per-read via x402, and get paid to challenge or vote. Every persona is a paid participant, not UI flavor.
- All of the above settle on **Mimir.sol** on Arc Testnet, with `evidenceHash = keccak256(raw evidence)` committed on-chain so any verdict is independently re-verifiable.

Stack outside Circle

Next.js 16 (App Router) + TypeScript + Tailwind, wagmi v3 / viem v2 for wallet and chain interaction, Solidity ^0.8.20 (`viaIR`) for the contract, Neon Postgres as an optional read-index cache, XMTP for encrypted creator/challenger chat, next-intl for i18n. Frontend on Vercel, long-running agent workers on Railway.

Proof, not pitch

Live contract on Arc Testnet: **181 claims created, 104 resolved** in native USDC (v1, immutable), ahead of the pull-payment-safety v2 upgrade currently live.